

```
In [9]: ▶ import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
import warnings
warnings.filterwarnings('ignore')
```

```
In [2]: ▶ df=pd.read_excel("Mall_Customers.xlsx")
```

```
In [4]: ▶ df.head()
```

Out[4]:

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Univariate Analysis

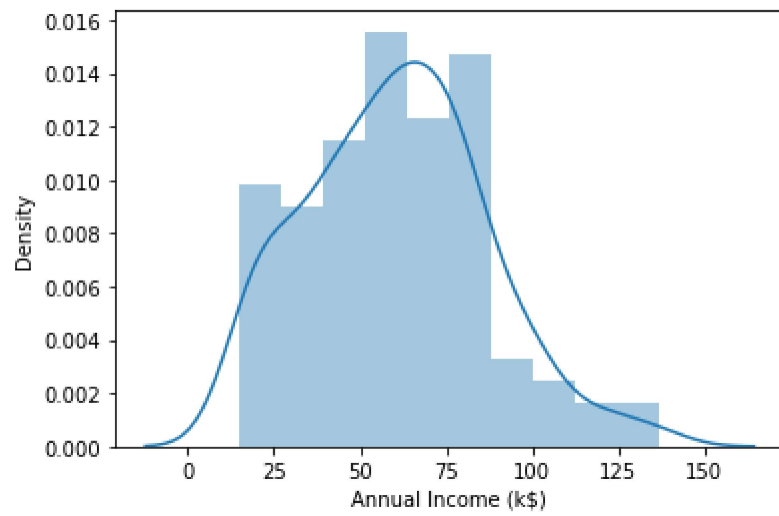
```
In [5]: ▶ df.describe()
```

Out[5]:

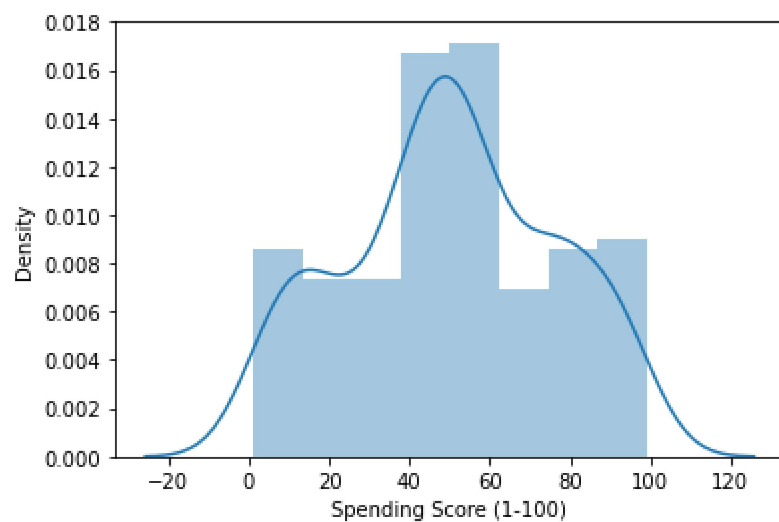
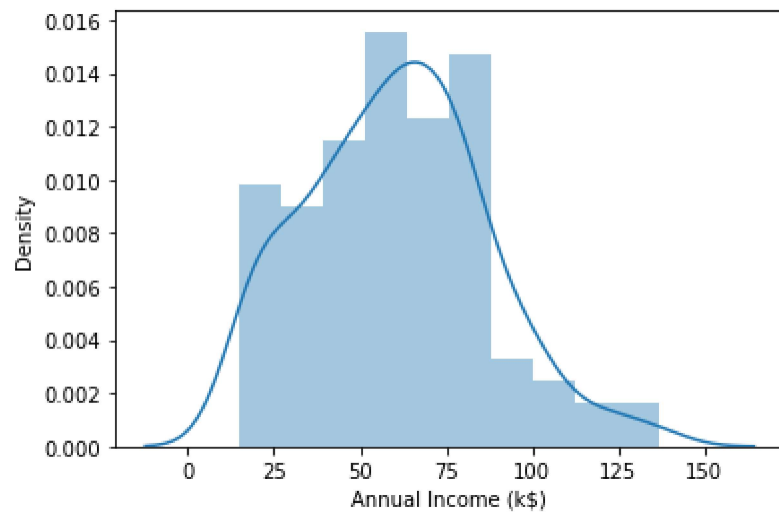
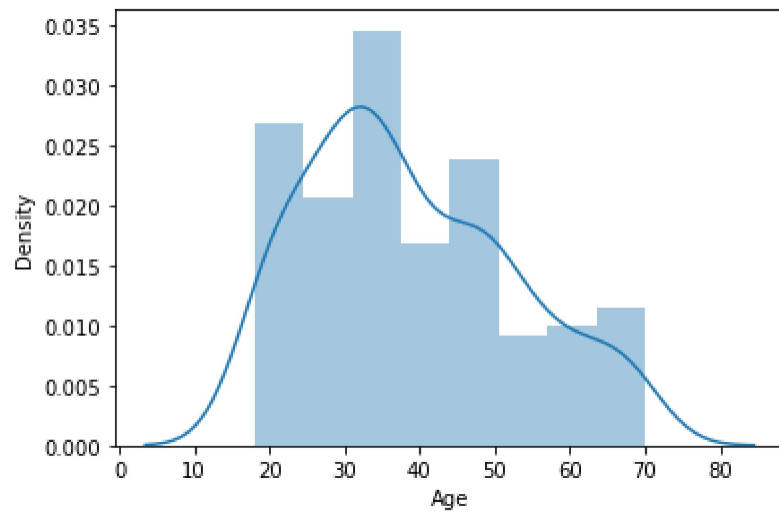
	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
In [10]: sns.distplot(df['Annual Income (k$)'])
```

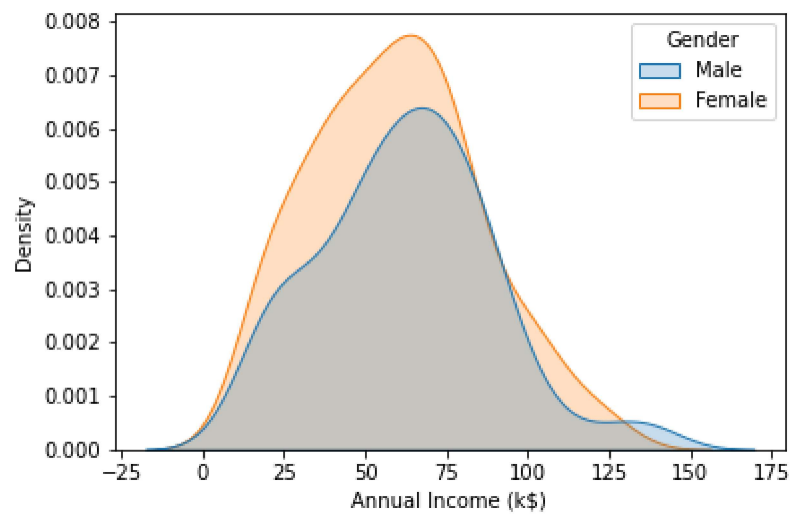
```
Out[10]: <AxesSubplot:xlabel='Annual Income (k$)', ylabel='Density'>
```



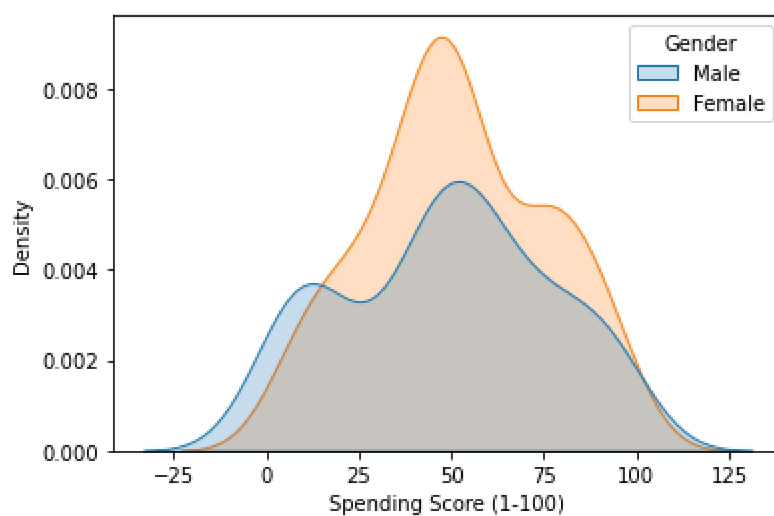
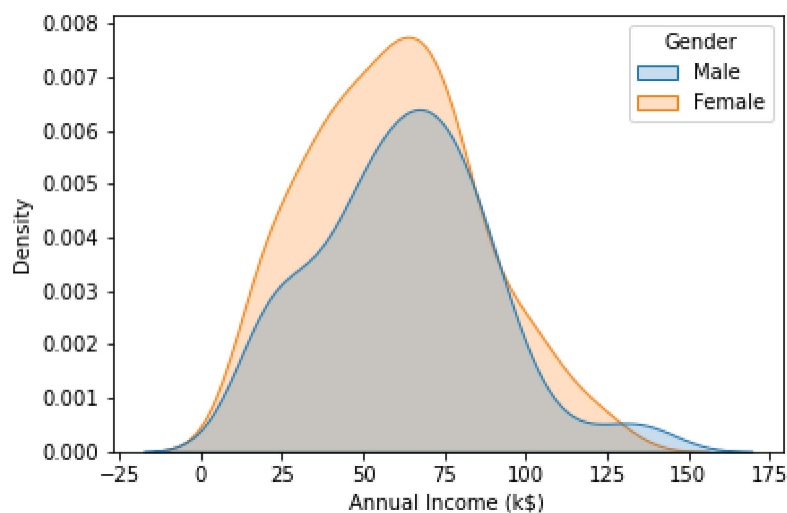
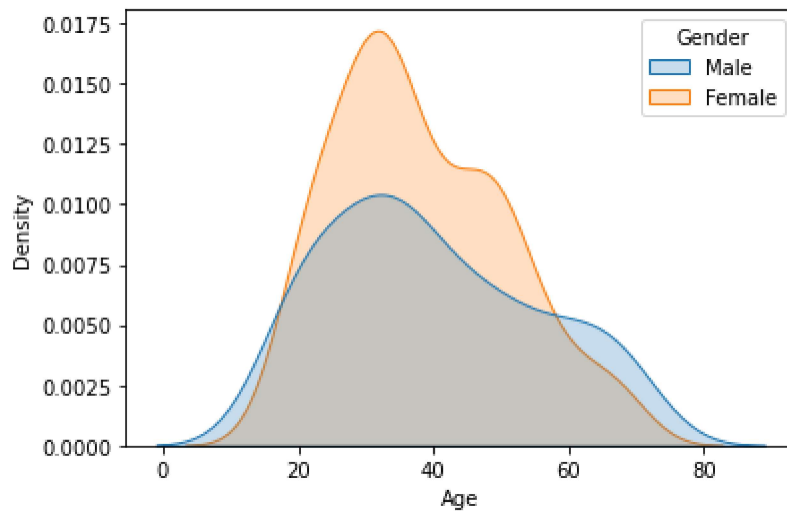
```
In [13]: columns=['Age', 'Annual Income (k$)', 'Spending Score (1-100)']  
for i in columns:  
    plt.figure()  
    sns.distplot(df[i])
```



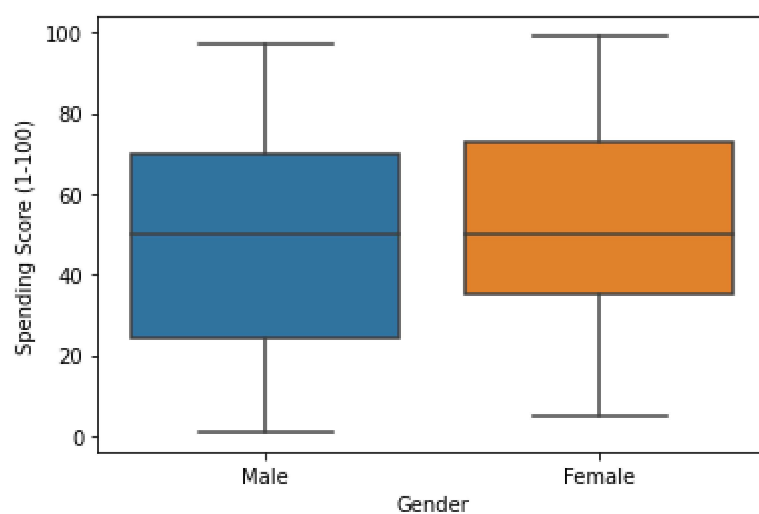
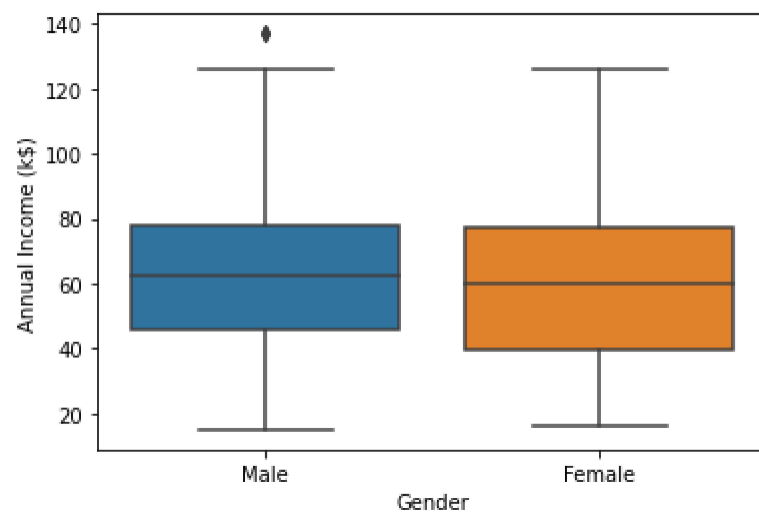
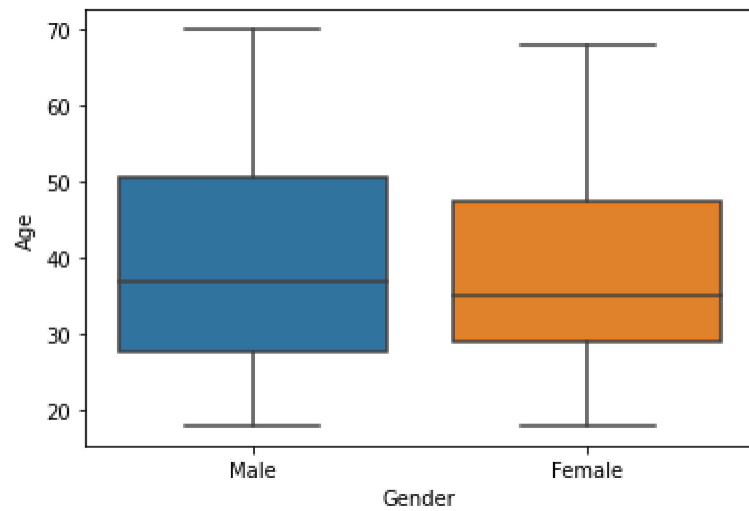
```
In [14]: sns.kdeplot(df['Annual Income (k$)'], shade=True, hue=df['Gender']);
```



```
In [15]: columns=['Age', 'Annual Income (k$)', 'Spending Score (1-100)']  
for i in columns:  
    plt.figure()  
    sns.kdeplot(df[i], shade=True, hue=df['Gender'])
```



```
In [16]: columns=['Age', 'Annual Income (k$)', 'Spending Score (1-100)']  
for i in columns:  
    plt.figure()  
    sns.boxplot(data=df, x='Gender', y=df[i])
```



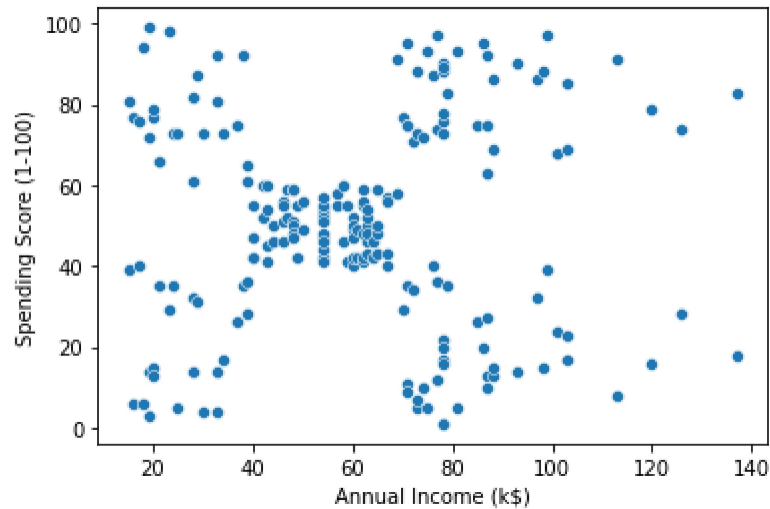
```
In [18]: df['Gender'].value_counts(normalize=True)
```

```
Out[18]: Female    0.56  
        Male      0.44  
        Name: Gender, dtype: float64
```

Bivariate Analysis

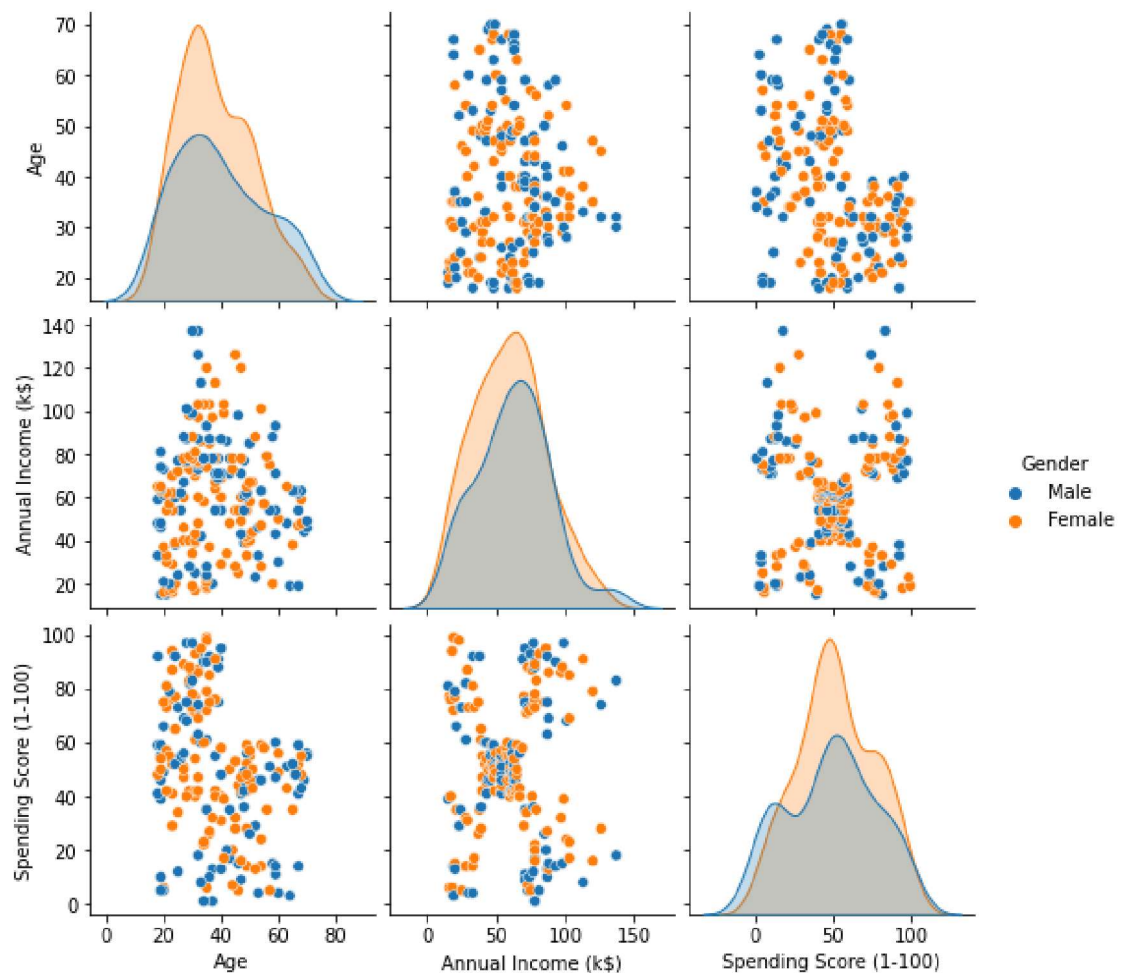
In [19]: `sns.scatterplot(data=df,x='Annual Income (k$)',y='Spending Score (1-100)')`

Out[19]: `<AxesSubplot:xlabel='Annual Income (k$)', ylabel='Spending Score (1-100)'\>`



In [23]: `df=df.drop('CustomerID',axis=1)
sns.pairplot(df,hue='Gender')`

Out[23]: `<seaborn.axisgrid.PairGrid at 0x165bcda8940>`



```
In [25]: df.groupby(['Gender'])['Age', 'Annual Income (k$)',
      'Spending Score (1-100)'].mean()
```

Out[25]:

	Age	Annual Income (k\$)	Spending Score (1-100)
Gender			
Female	38.098214	59.250000	51.526786
Male	39.806818	62.227273	48.511364

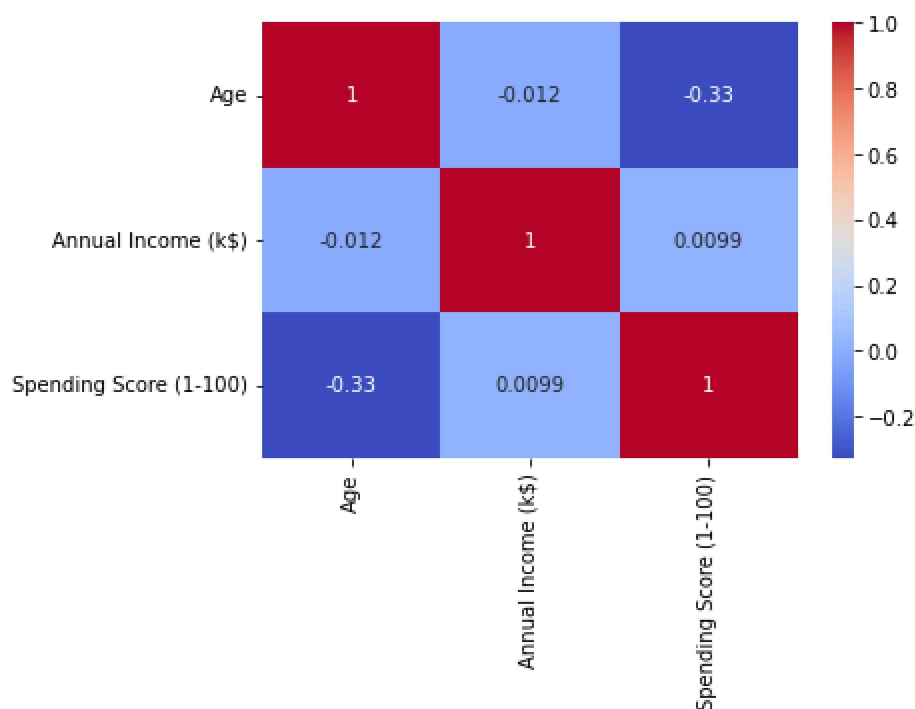
```
In [26]: df.corr()
```

Out[26]:

	Age	Annual Income (k\$)	Spending Score (1-100)
Age	1.000000	-0.012398	-0.327227
Annual Income (k\$)	-0.012398	1.000000	0.009903
Spending Score (1-100)	-0.327227	0.009903	1.000000

```
In [27]: sns.heatmap(df.corr(),annot=True,cmap='coolwarm')
```

Out[27]: <AxesSubplot:>



Clustering - Univariate, Bivariate, Multivariate

```
In [28]: clustering1=KMeans()
```

```
In [30]: clustering1.fit(df[['Annual Income (k$)']])
```

Out[30]: KMeans()


```
clustering1.labels_
```

```
array([4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,
       4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 6, 6, 6, 6, 6, 6, 6, 6,
       6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 2, 2,
       2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
       2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,
       5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,
       5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 3, 3, 3, 3,
       3, 3])
```

```
df['Income Cluster']=clustering1.labels_  
df.head()
```

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Income Cluster
0	Male	19	15	39	4
1	Male	21	15	81	4
2	Female	20	16	6	4
3	Female	23	16	77	4
4	Female	31	17	40	4

```
df['Income Cluster'].value_counts()
```

```
5    42
2    36
4    32
6    28
0    26
1    16
7    14
3     6
Name: Income Cluster, dtype: int64
```

```
clustering1.inertia_
```

2822.4996947496948

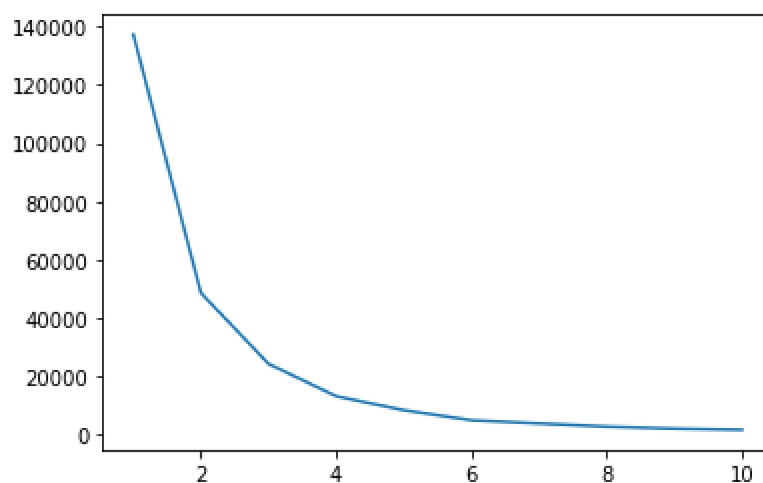
```
intertia_scores=[]
for i in range(1,11):
    kmeans=KMeans(n_clusters=i)
    kmeans.fit(df[['Annual Income (k$)']])
    inertia_scores.append(kmeans.inertia )
```

```
In [40]: ▶ inertia_scores
```

```
Out[40]: [137277.28,  
         48660.88888888889,  
         24361.259213759215,  
         13278.112713472485,  
         8481.496190476191,  
         5050.904761904762,  
         3931.9880952380954,  
         2862.250427350428,  
         2173.287445887446,  
         1758.1453823953825]
```

```
In [41]: ▶ plt.plot(range(1,11),inertia_scores)
```

```
Out[41]: [<matplotlib.lines.Line2D at 0x165be5f76d0>]
```



```
In [42]: ▶ df.groupby('Income Cluster')['Age', 'Annual Income (k$)',  
        'Spending Score (1-100)'].mean()
```

```
Out[42]:
```

	Age	Annual Income (k\$)	Spending Score (1-100)
Income Cluster			
0	39.846154	38.846154	49.730769
1	39.375000	87.625000	48.937500
2	41.055556	62.111111	49.000000
3	36.833333	127.666667	49.666667
4	34.906250	22.000000	49.656250
5	35.428571	75.095238	51.095238
6	46.035714	50.428571	50.892857
7	36.500000	102.000000	53.000000

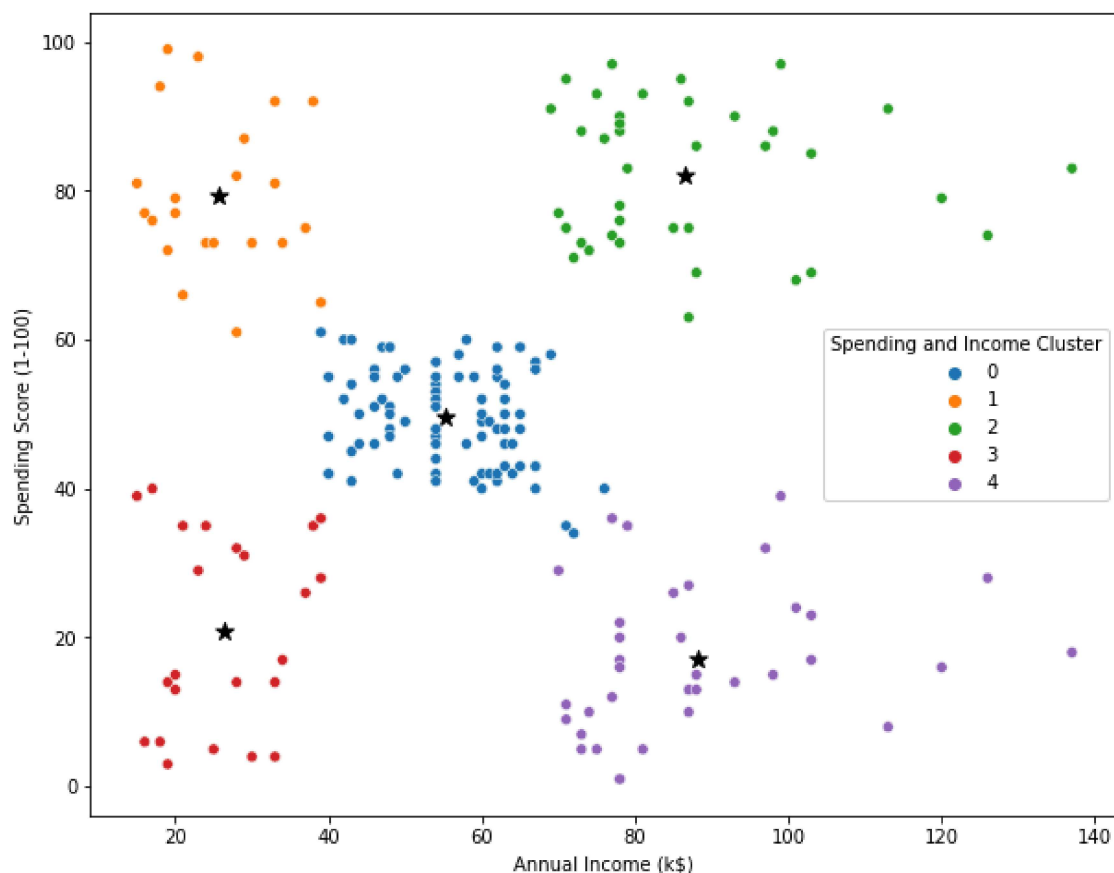
```
In [43]: #Bivariate Clustering
clustering2=KMeans(n_clusters=5)
clustering2.fit(df[['Annual Income (k$)','Spending Score (1-100)']])
df['Spending and Income Cluster']=clustering2.labels_
df.head()
```

Out[43]:

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Income Cluster	Spending and Income Cluster
0	Male	19	15	39	4	3
1	Male	21	15	81	4	1
2	Female	20	16	6	4	3
3	Female	23	16	77	4	1
4	Female	31	17	40	4	3

```
In [44]: centers=pd.DataFrame(clustering2.cluster_centers_)
centers.columns=['x','y']
```

```
In [59]: plt.figure(figsize=(10,8))
plt.scatter(x=centers['x'],y=centers['y'],s=100,c='black',marker='*')
sns.scatterplot(data=df,x='Annual Income (k$)',y='Spending Score (1-100)',
plt.savefig('clustering_bivariate.png')
```



```
In [46]: ▶ pd.crosstab(df['Spending and Income Cluster'],df['Gender'],normalize='inde
```

Out[46]:

	Gender	Female	Male
Spending and Income Cluster			
0		0.592593	0.407407
1		0.590909	0.409091
2		0.538462	0.461538
3		0.608696	0.391304
4		0.457143	0.542857

```
In [47]: ▶ df.groupby('Spending and Income Cluster')['Age','Annual Income (k$)',
           'Spending Score (1-100)'].mean()
```

Out[47]:

	Age	Annual Income (k\$)	Spending Score (1-100)
Spending and Income Cluster			
0	42.716049	55.296296	49.518519
1	25.272727	25.727273	79.363636
2	32.692308	86.538462	82.128205
3	45.217391	26.304348	20.913043
4	41.114286	88.200000	17.114286

```
In [48]: ▶ #multivariate clustering
           from sklearn.preprocessing import StandardScaler
```

```
In [49]: ▶ scale=StandardScaler()
```

```
In [50]: ▶ dff=pd.get_dummies(df,drop_first=True)
           df.head()
```

Out[50]:

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Income Cluster	Spending and Income Cluster
0	Male	19	15	39	4	3
1	Male	21	15	81	4	1
2	Female	20	16	6	4	3
3	Female	23	16	77	4	1
4	Female	31	17	40	4	3

```
In [51]: ▶ dff.columns
```

Out[51]: Index(['Age', 'Annual Income (k\$)', 'Spending Score (1-100)', 'Income Cluster',
 'Spending and Income Cluster', 'Gender_Male'],
 dtype='object')

```
In [52]: dff=dff[['Age', 'Annual Income (k$)', 'Spending Score (1-100)', 'Gender_Male']
dff.head()
```

Out[52]:

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Income Cluster	Spending and Income Cluster
0	Male	19	15	39	4	3
1	Male	21	15	81	4	1
2	Female	20	16	6	4	3
3	Female	23	16	77	4	1
4	Female	31	17	40	4	3

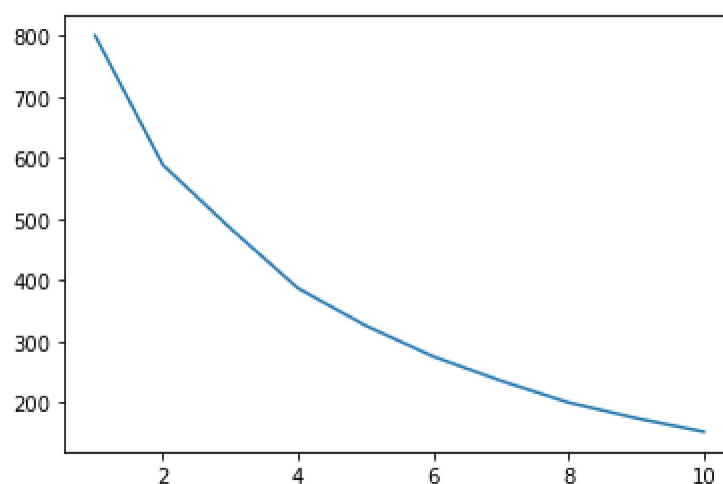
```
In [54]: dff=scale.fit_transform(dff)
dff=pd.DataFrame(scale.fit_transform(dff))
dff.head()
```

Out[54]:

	0	1	2	3
0	-1.424569	-1.738999	-0.434801	1.128152
1	-1.281035	-1.738999	1.195704	1.128152
2	-1.352802	-1.700830	-1.715913	-0.886405
3	-1.137502	-1.700830	1.040418	-0.886405
4	-0.563369	-1.662660	-0.395980	-0.886405

```
In [57]: inertia_scores3=[]
for i in range(1,11):
    kmeans3=KMeans(n_clusters=i)
    kmeans3.fit(dff)
    inertia_scores3.append(kmeans3.inertia_)
plt.plot(range(1,11),inertia_scores3)
```

Out[57]: [<matplotlib.lines.Line2D at 0x165c2ac2a90>]



In [58]: `df`

Out[58]:

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Income Cluster	Spending and Income Cluster
0	Male	19	15	39	4	3
1	Male	21	15	81	4	1
2	Female	20	16	6	4	3
3	Female	23	16	77	4	1
4	Female	31	17	40	4	3
...	...	...	...	...	...	...
195	Female	35	120	79	3	2
196	Female	45	126	28	3	4
197	Male	32	126	74	3	2
198	Male	32	137	18	3	4
199	Male	30	137	83	3	2

200 rows × 6 columns

In [60]: `df.to_csv('Clustering.csv')`